



DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

VI Sem Minor Project

24EEEW301

Project Title: An adaptive Kalman filter based approach for SOC estimation of Lithium-ion cell using Coulomb Counting and Open Circuit Voltage.

by

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CERTIFICATE

This is to certify that the project entitled “An adaptive Kalman filter based approach for SOC estimation of Lithium-ion cell using Coulomb Counting and Open Circuit Voltage” submitted to KLE Technological University Dr. M S Sheshgiri campus belagavi for the partial fulfillment of the requirement for the course of Minor project (24EEEW301) by Abhishek Rolli (02FE23BEE042), Aditya Patil (02FE23BEE056), Koustubh Bandiwadekar (02FE23BEE045) and Shrigouri Chikkamath (02FE23BEE010) students in the department of Electrical and Electronics Engineering, KLE Technological University Dr. M S Sheshgiri campus belagavi, is a Bonafide record of the work carried out by them under my guidance. The contents of this report, in full or in parts, have not been submitted to any other institute or university for the award of any other course completion.

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Abstract

State of Charge (SOC) estimation plays a vital role in ensuring the efficient and safe operation of Lithium-ion batteries used in Electric Vehicles (EVs). Traditional SOC estimation methods such as Open Circuit Voltage (OCV) and Coulomb Counting suffer from limitations including the need for battery rest conditions, error accumulation, and reduced accuracy under dynamic operating conditions. To address these challenges, this project proposes a real-time SOC estimation system based on the Extended Kalman Filter (EKF).

The proposed system utilizes battery voltage and current measurements along with a battery equivalent circuit model to estimate SOC more accurately. hardware implementation for real-time monitoring. The EKF effectively handles the nonlinear behaviour of Lithium-ion batteries and reduces the impact of measurement noise and model uncertainties.

The developed system provides improved estimation accuracy, reliability, and real-time performance, making it suitable for Battery Management Systems (BMS) in Electric Vehicle applications.

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1. Introduction:

1.1 Problem statement:

The existing Variational EKF-based SOC estimation method improves estimation accuracy for lithium-ion batteries but is limited mainly to software-level validation and remains sensitive to initialization errors. There is a need to develop a real-time hardware-based SOC estimation system using Adaptive Kalman Filter to improve practical implementation, reduce initialization sensitivity, and enhance the reliability of Battery Management Systems.

1.2 Real-World Context of the selected use case

Electric vehicles are becoming increasingly popular, making efficient battery management more important than ever. Since the battery is the heart of an EV, accurately knowing its State of Charge (SOC) helps improve vehicle range, battery life, and safety. However, real-world driving conditions cause battery behavior to change continuously, making SOC estimation challenging. This project focuses on developing a reliable real-time SOC estimation system using Kalman Filter techniques to enhance the performance and effectiveness of Battery Management Systems in EVs.

2. Literature Survey & Identified gaps:

2.1 Literature survey:

[1] Lee and Won (2023) proposed an Enhanced Coulomb Counting (ECC) method for simultaneous SOC and State of Health (SOH) estimation. The method combines Open Circuit Voltage (OCV), coulombic efficiency, and internal resistance information to reduce errors caused by battery aging. Experimental results showed improved estimation accuracy compared to conventional Coulomb Counting and Extended Kalman Filter (EKF) methods .

[2] Ziheng Zhou et al. (2023) developed a Variational Extended Kalman Filter (VEKF) for SOC estimation. The proposed approach continuously updates battery model parameters, thereby reducing estimation errors and improving performance under varying operating conditions. The study demonstrated better accuracy than traditional Coulomb Counting techniques.

[3] Yunsheng Fan et al. (2025) introduced an enhanced OCV-based SOC estimation model that considers internal resistance and real-time current effects. Unlike conventional OCV methods, the proposed model does not require long battery rest periods and provides more accurate SOC estimation during practical battery operation

[4] Teodor-Iulian Voicila et al. (2025) presented an improved OCV estimation technique for LiFePO₄ batteries by integrating OCV measurements with an equivalent circuit battery model. The method effectively captures battery dynamic behavior and improves SOC estimation accuracy under real-world conditions.

[5] Muhammad Hamza et al. (2023) proposed a hybrid deep learning-based SOC estimation model that utilizes battery voltage, current, temperature, and SOC data for training. Although the model achieved high estimation accuracy, it requires significant computational resources and memory, making hardware implementation on low-cost microcontrollers challenging.

[6] Na Shi et al. (2022) presented an Adaptive Extended Kalman Filter (AEKF) based SOC estimation method with improved parameter identification. The adaptive mechanism dynamically updates filter parameters, resulting in improved estimation accuracy and robustness under changing battery operating conditions

[7] Hasanur Zman Anonto (2025) presented a comparative analysis of two widely used battery State-of-Charge (SOC) estimation techniques for electric vehicles: the Coulomb Counting (CC) method and the Kalman Filter (KF) approach. The study evaluates their performance in terms of estimation accuracy, computational complexity, and robustness under varying operating conditions. Results indicate that the Coulomb Counting method is simple and computationally efficient but suffers from cumulative measurement errors and sensitivity to initial SOC values. In contrast, the Kalman Filter provides more accurate and reliable SOC estimation by continuously correcting prediction errors using real-time measurements. The research concludes that Kalman Filter-based estimation is better suited for advanced Battery Management Systems (BMS) in electric vehicles, where high accuracy and adaptability are essential

[8] A. Manthopoulos (2020) reviewed and compared various lithium-ion battery State-of-Charge (SOC) estimation methods for electric vehicles, including Coulomb Counting, OCV-based, model-based, and data-driven techniques. The authors evaluated these methods based on accuracy, computational complexity, and robustness. Their analysis showed that advanced approaches such as Kalman Filters and machine learning algorithms provide more accurate SOC estimation than conventional methods. The study concluded that hybrid estimation techniques offer the best balance between accuracy, reliability, and real-time implementation in Battery Management Systems.

[9] Ko et al. (2022) proposed a novel lithium-ion battery capacity estimation method that combines an enhanced Coulomb Counting technique with Kalman Filtering to improve estimation accuracy. The approach compensates for the limitations of traditional Coulomb Counting by reducing cumulative measurement errors and correcting estimation drift. Experimental validation demonstrated improved capacity estimation performance under varying operating conditions. The authors concluded that the proposed method offers a reliable and practical solution for battery monitoring and Battery Management System (BMS) applications.

[10] Haoyu Li (2025) reviewed recent advancements in artificial intelligence (AI)-based battery state estimation for electric vehicles, focusing on the transition from research innovations to practical applications. The study analyzed various AI techniques, including machine learning and deep learning models, for estimating battery states such as SOC and SOH. The authors highlighted challenges related to data quality, model interpretability, computational requirements, and real-time implementation. The paper concluded that integrating AI-driven estimation methods with Battery Management Systems can significantly enhance battery performance, reliability, and operational efficiency in real-world electric vehicle applications.

2.2 Identified Gaps:

From the literature survey, it was observed that most SOC estimation techniques have been validated primarily through simulation and software-based models, with limited emphasis on real-time hardware implementation. In addition, many recent approaches employ complex algorithms such as deep learning methods, which increase computational requirements and make practical implementation on low-cost embedded systems challenging. Therefore, there is a need for a simple, cost-effective, and hardware-validated SOC estimation approach that can be implemented in real-time battery monitoring applications. This project addresses these gaps by developing and testing the SOC estimation method through both simulation and hardware implementation.

3. Objectives:

- To survey the working principles of Lithium-ion batteries and the importance of State of Charge (SOC) estimation in Battery Management Systems (BMS).
- To analyze the limitations of conventional SOC estimation methods such as Open Circuit Voltage (OCV) and Coulomb Counting.
- To implement an Extended Kalman Filter (EKF) algorithm for accurate SOC estimation under varying operating conditions.
- To simulate and validate the proposed SOC estimation technique using a micro-controller.
- To design and implement a hardware prototype for real-time battery voltage and current monitoring.
- To evaluate the performance, accuracy, and reliability of the proposed method for Electric Vehicle battery applications.

4. Methodology:

The proposed system estimates the State of Charge (SOC) of a Lithium-ion battery using an Extended Kalman Filter (EKF) implemented on an ESP32 microcontroller. The methodology combines Coulomb Counting and real-time voltage measurements to improve SOC estimation accuracy.

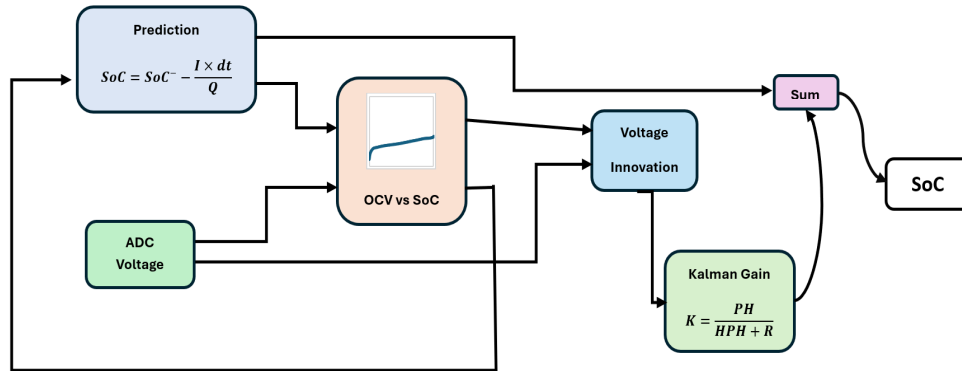


Fig (1): Used Methodology

4.1 Prerequisite:

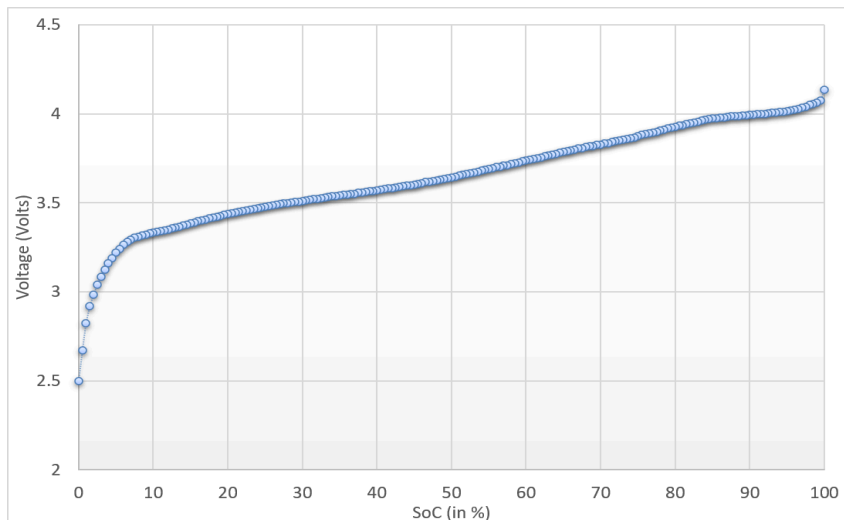


Fig (2) : OCV vs SoC curve

Fig (2) shows the experimentally obtained OCV–SOC characteristic curve of the Li-ion cell derived from the 0.2C discharge test data. The curve exhibits the typical battery behavior with a steep voltage variation at low and high SOC regions and a relatively flat plateau in the mid-SOC range. This OCV–SOC relationship was used to generate the lookup table required for SOC estimation in the EKF algorithm.

4.2 SOC Prediction using Coulomb Counting

The first stage of EKF is the prediction step. SOC is predicted using the Coulomb Counting method as shown in equation(1) , which calculates the charge removed from the battery over time.

$$SoC = SoC^- + \frac{I \times dt}{Q} \quad (1)$$

Where,

SoC = Updated State of Charge of the battery (%)

SoC⁻ = Predicted State of Charge from the previous estimation step (%)

I = Battery current (A or mA)

dt = Time interval between consecutive measurements (h)

Q = Nominal battery capacity (Ah or mAh)

SOC is estimated based on battery current, sampling interval, and battery capacity. This predicted SOC serves as the initial estimate for the EKF algorithm.

However, Coulomb Counting alone accumulates errors over time due to sensor noise and inaccuracies in current measurement.

4.3 Voltage-Based Measurement

The battery voltage is measured by the ESP32 ADC through a voltage divider consisting of two 10 kΩ resistors, which reduces the voltage to half its value and keeps it within the 3.3 V input limit of the ESP32. The 10 kΩ resistors were selected to minimize current consumption and power loss, unlike lower resistance values which would draw unnecessary current.

The measured voltage is then compared with the predicted voltage corresponding to the estimated SOC. The resulting error is used by the Extended Kalman Filter (EKF) to correct the SOC estimate and improve its accuracy.

The voltage measurement acts as a correction source to compensate for the drift and accumulated errors of the Coulomb Counting method.

4.4 EKF Error Correction

In the correction stage, the Extended Kalman Filter updates the predicted SOC using the difference between the measured battery voltage and the predicted battery voltage.

$$K = \frac{PH}{HPH+R} \quad (2)$$

Where,

K = Kalman Gain

P = Error covariance matrix representing the uncertainty in the predicted state estimate

H = Measurement matrix (or Jacobian matrix in EKF) relating the state variable to the measured output

R = Measurement noise covariance representing the uncertainty in the sensor measurements

From equation (2) the Kalman Gain determines how much confidence should be given to the measured data versus the predicted estimate. Based on this gain, the SOC estimate is corrected and updated.

This process significantly reduces estimation errors and improves SOC accuracy under varying operating conditions.

4.5 Estimation of Noise Parameters

a. Measurement Noise Covariance(R):

The measurement noise covariance (R) was obtained from the variance of 100 battery voltage samples recorded using the ESP32 ADC under no-load conditions.

$$R = \text{Variance}(V_{ADC}) \quad (3)$$

Where,

V_{adc} = Battery voltage measured using the ESP32 ADC

b. Process Noise Covariance (Q)

The process noise covariance (Q) was calculated from the variance of the prediction error between the OCV-based SOC estimate and the coulomb-counting SOC estimate over 100 samples.

$$Q = \text{Variance}(\text{SoC}_{ADC} - \text{SoC}_{CC}) \quad (4)$$

Where,

SoC_{ADC} = SoC estimated using the OCV–SoC lookup table

SoC_{CC} = SOC estimated using the coulomb counting method

c. Error Covariance (P)

The error covariance (P) represents the uncertainty in the SOC estimate. During the prediction step, it is updated using the process noise covariance as

$$P_k^- = P_{k-1} + Q \quad (5)$$

and after measurement correction,

$$P_k = (1 - KH) \times P_k^- \quad (6)$$

Where,

P_k^- = Predicted error covariance

P_{k-1} = Previous error covariance

P_k = Updated error covariance

d. Measurement Matrix (H)

The measurement matrix (**H**) is the Jacobian of the OCV–SOC relationship and represents the sensitivity of battery voltage to changes in SOC.

$$H = \frac{d(OCV)}{d(SoC)} \quad (7)$$

It is approximated using consecutive points from the OCV–SOC lookup table as

$$H = \frac{OCV_{i+1} - OCV_i}{SoC_{i+1} - SoC_i} \quad (8)$$

and is used in the computation of the Kalman Gain.

Where,

OCV_i, OCV_{i+1} = Consecutive Open Circuit Voltage values

SoC_i, SoC_{i+1} = Corresponding State of Charge values

4.6 Real-Time SOC Estimation

The prediction and correction steps are executed repeatedly in real time. From equation (3) the EKF continuously updates the SOC estimate whenever new voltage measurements become available. This enables accurate battery monitoring while minimizing the effects of sensor noise, modeling errors, and operating condition variations.

$$SoC = SoC_k^- + K * (innovation) \quad (9)$$

Where,

SoC_k^- = Predicted SoC (Coulomb Counting

$$innovation = V_{Measured} - V_{Predicted} \quad (10)$$

$V_{measured}$ = ADC measured Voltage

$V_{\text{predicted}}$ = Voltage predicted by the SoC by putting the predicted SoC value back to the OCV vs SoC curve

4.7 Role of Kalman Gain (K):

Kalman gain determines how much the estimated SOC should be corrected using the measured battery voltage .

- **High K** : More trust in the voltage measurement
- **Low K** : More trust in the prediction (coulomb counting).

4.8 Code Flowchart

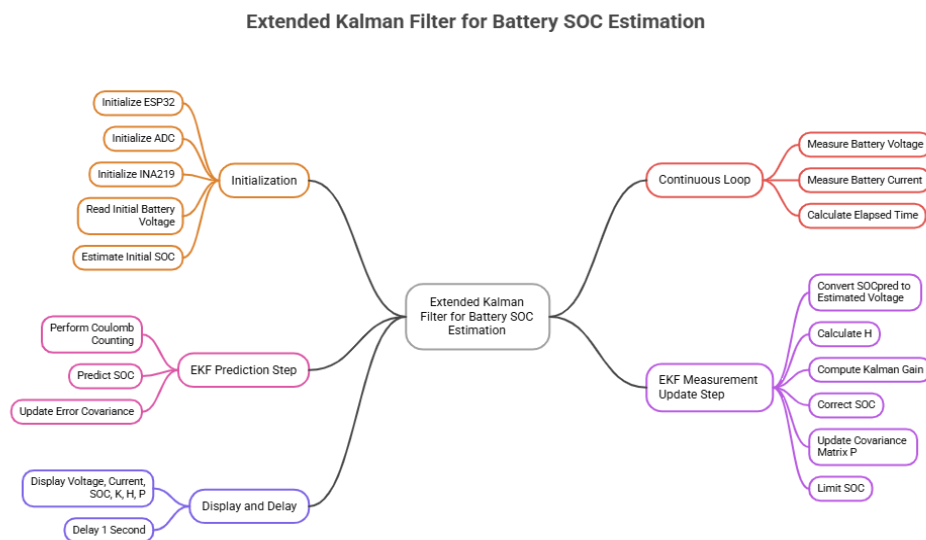


Fig (3) Code Flowchart

Fig (3) illustrates the workflow of the Extended Kalman Filter (EKF) implemented for battery State of Charge (SOC) estimation. The process begins with system initialization and initial SOC estimation, followed by a continuous loop consisting of voltage/current measurement, coulomb-counting based SOC prediction, and EKF measurement update using the battery voltage model. Finally, the estimated SOC, voltage, current, Kalman gain, and covariance values are displayed for real-time monitoring.

5. Hardware Implementation:

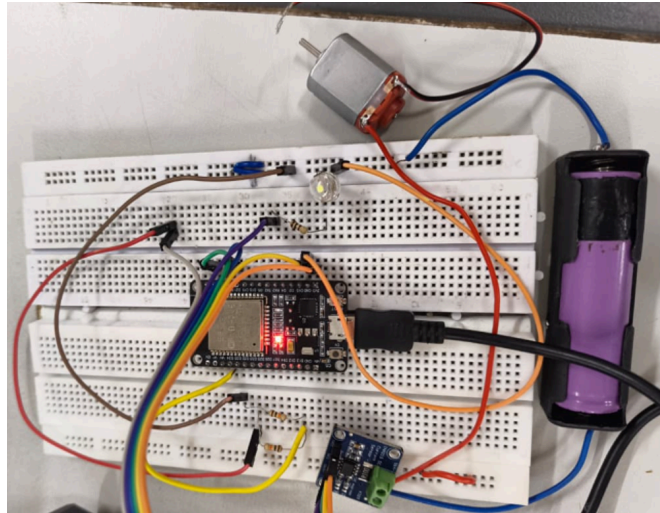


Fig (4):Hardware implementation

Fig (4) shows the experimental hardware setup developed for real-time battery SOC estimation. The setup consists of an ESP32 microcontroller, INA219 current sensor module, Li-ion battery, DC motor load, and supporting voltage sensing circuitry assembled on a breadboard. The measured battery parameters are processed by the EKF algorithm implemented on the ESP32 to estimate the State of Charge (SOC) in real time.

5.1 Load Analysis:

The Device was tested under three load conditions.

a. NO load:

At no load the output connection was open circuited and the data was taken from the serial monitor.

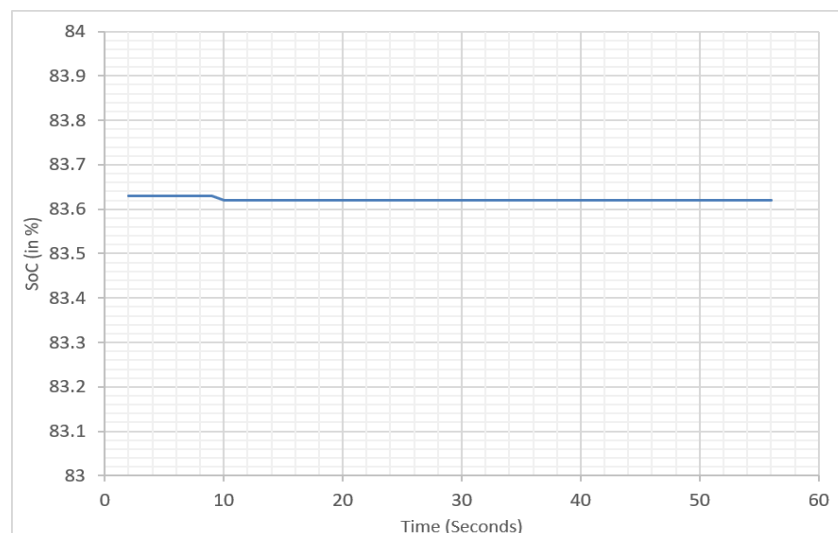


Fig (5):Time vs SoC (NO load)

Fig (5) shows the estimated State of Charge (SOC) of the battery under no-load conditions. Since there is negligible current flow, the battery experiences almost no charge depletion, resulting in a nearly constant SOC throughout the observation period. The stable SOC profile also indicates that the EKF-based estimation algorithm is operating consistently under steady-state conditions.

b. LED load:

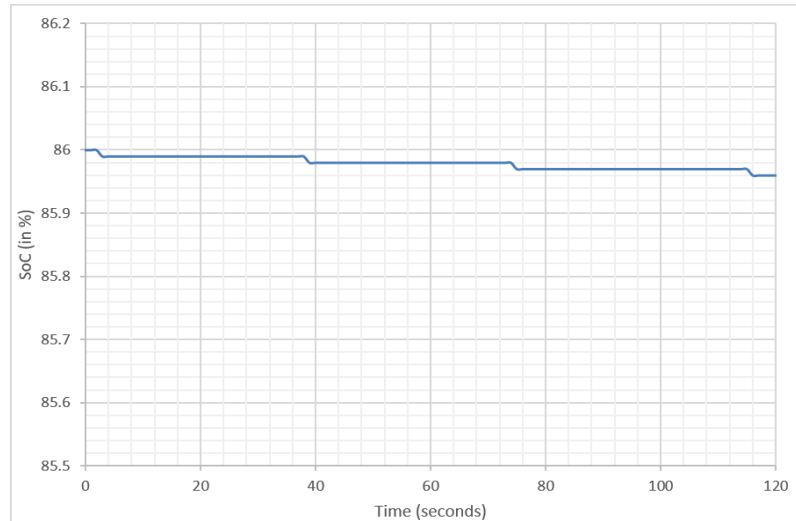


Fig (6): Time vs SoC (LED load)

Fig (6) illustrates the variation of battery State of Charge (SOC) when supplying a low-power LED load connected in series with a $100\ \Omega$ current-limiting resistor. The load draws an average current of approximately 12 mA, resulting in a gradual discharge of the battery and a corresponding decrease in SOC over time. The smooth SOC profile demonstrates the ability of the EKF algorithm to provide stable and reliable SOC estimation under light-load operating conditions.

c. Motor load:

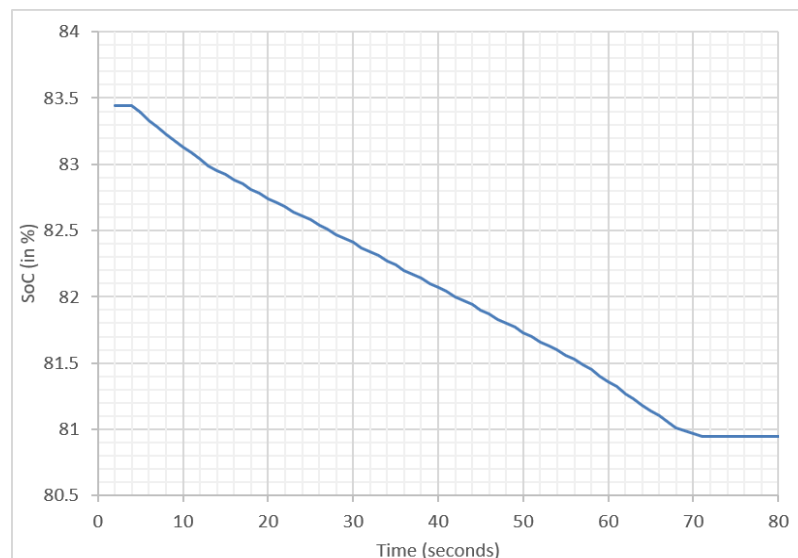


Fig (7): Time vs SoC (Motor load)

Fig (7) shows the variation of battery State of Charge (SOC) when supplying a DC Motor Load. A small brushed DC hobby motor was used as a dynamic electromechanical load, drawing a peak current of approximately 1 A from the 2600 mAh Li-ion battery. Compared to the LED load, the SOC decreases at a faster rate due to the significantly higher current demand. The steeper SOC slope confirms the increased discharge rate, while the EKF continues to provide smooth and stable SOC estimation under high-current operating conditions.

6. Results and Discussion:

The proposed State of Charge (SOC) estimation system was successfully implemented using an ESP32 microcontroller, INA219 current sensor, voltage divider circuit, with Extended Kalman Filter (EKF) algorithm. Battery voltage and current were measured in real time and processed by the EKF to estimate SOC accurately.

An Open Circuit Voltage (OCV) versus SOC characteristic curve was generated for the Lithium-ion battery of 2600mah. The obtained curve shows a gradual increase in battery voltage from approximately 2.5 V at 0% SOC to 4.1 V at 100% SOC. The curve follows the expected nonlinear behavior of Lithium-ion batteries, validating the battery model used in the project.

The EKF-based approach effectively combines Coulomb Counting predictions with voltage measurements to reduce estimation errors. Compared to conventional Coulomb Counting, the proposed method minimizes error accumulation and provides more reliable SOC estimation under varying operating conditions. The use of experimentally determined process noise (Q) and measurement noise (R) parameters further improved estimation accuracy and filter stability.

The hardware implementation demonstrated successful real-time monitoring and SOC estimation, indicating the feasibility of using the proposed system in Battery Management System (BMS) applications for Electric Vehicles.

7. Conclusion

This project presented the design and implementation of a real-time State of Charge (SOC) estimation system for Lithium-ion batteries using the Extended Kalman Filter (EKF). The proposed method combines Coulomb Counting with voltage-based correction to overcome the limitations of conventional SOC estimation techniques.

The system was successfully implemented using an ESP32 microcontroller, INA219 current sensor, and voltage measurement through an ADC-based voltage divider circuit. Experimental results showed that the EKF effectively reduces estimation errors caused by sensor noise and drift while providing reliable SOC estimation in real time.

The generated OCV-SOC characteristics closely matched the expected behaviour of Lithium-ion batteries, validating the effectiveness of the battery model and estimation approach. Overall, the project demonstrates that EKF-based SOC estimation is a practical and accurate solution for Battery Management Systems in Electric Vehicle applications.

7.1 Future Scope:

- **Integration with Battery Management System (BMS)**

The proposed SOC estimation method can be integrated into a complete BMS for monitoring and protecting EV batteries in real time.

- **Simultaneous SOC and SOH Estimation**

Future work can focus on estimating both State of Charge (SOC) and State of Health (SOH) together to improve battery performance and lifespan prediction.

- **Wireless Battery Monitoring**

The system can be enhanced by incorporating IoT or wireless communication technologies for remote battery monitoring and data analysis.

- **Application to Different Battery Types**

The proposed method can be tested and adapted for various battery chemistries such as Lithium-ion, LiFePO_4 , and solid-state batteries.

- **Use of Artificial Intelligence Techniques**

Machine Learning and Deep Learning algorithms can be incorporated in the future to further improve SOC estimation accuracy under varying operating conditions.

- **Implementation of Adaptive Extended Kalman Filter (AEKF) for improved accuracy.**

- **Variation in Temperature:** Testing under varying temperature and load conditions.

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